

Forge — Application Synopsis & Use Cases

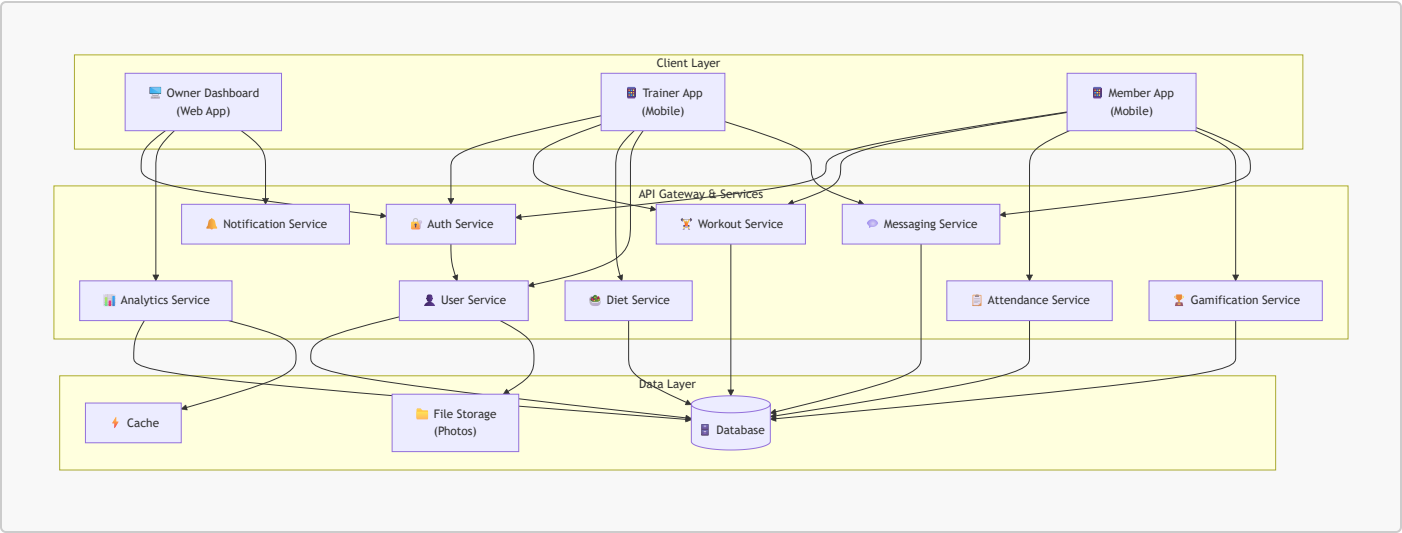
1. Executive Summary

Forge is a role-based gym management platform designed to streamline gym operations for **Owners**, empower **Trainers** to deliver personalized fitness programs, and keep **Members** engaged through gamified workout tracking. The app follows a minimalistic design philosophy — every feature is purposeful, every interaction is fast, and every insight is actionable.

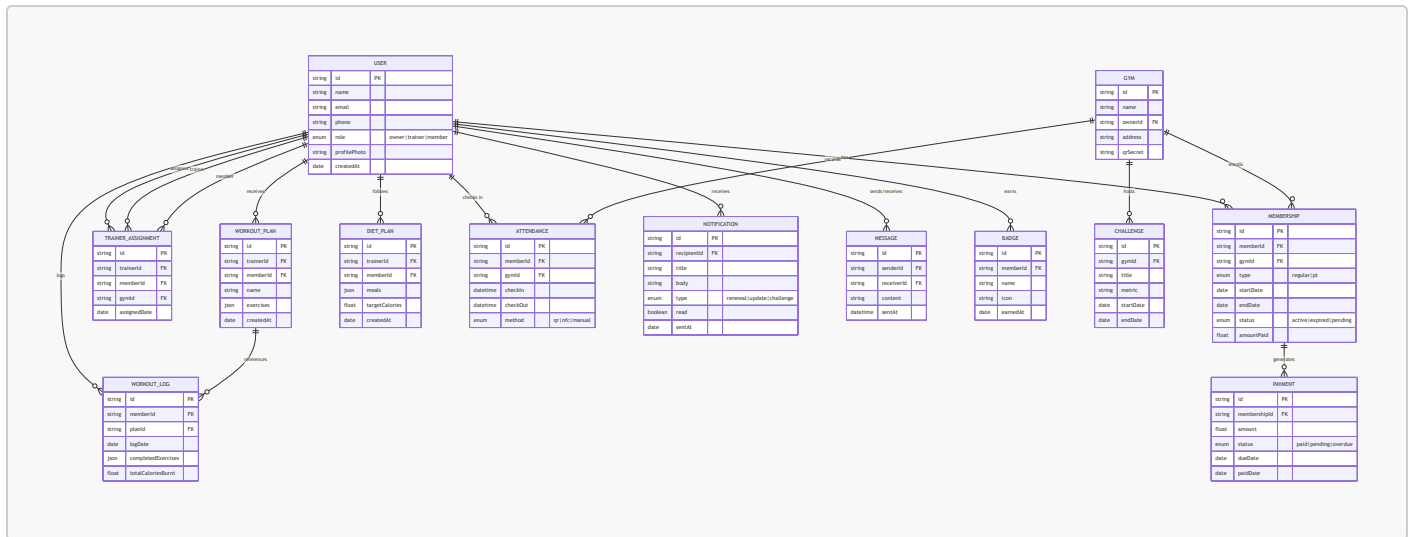
Target Users

Role	Who They Are	Primary Need
Owner	Gym founders, managers, franchise operators	Business visibility, revenue control, member retention
Trainer	Personal trainers, group class instructors	Client management, plan customization, progress tracking
Member	Gym-goers (regular & PT clients)	Workout tracking, motivation, attendance convenience

2. System Architecture

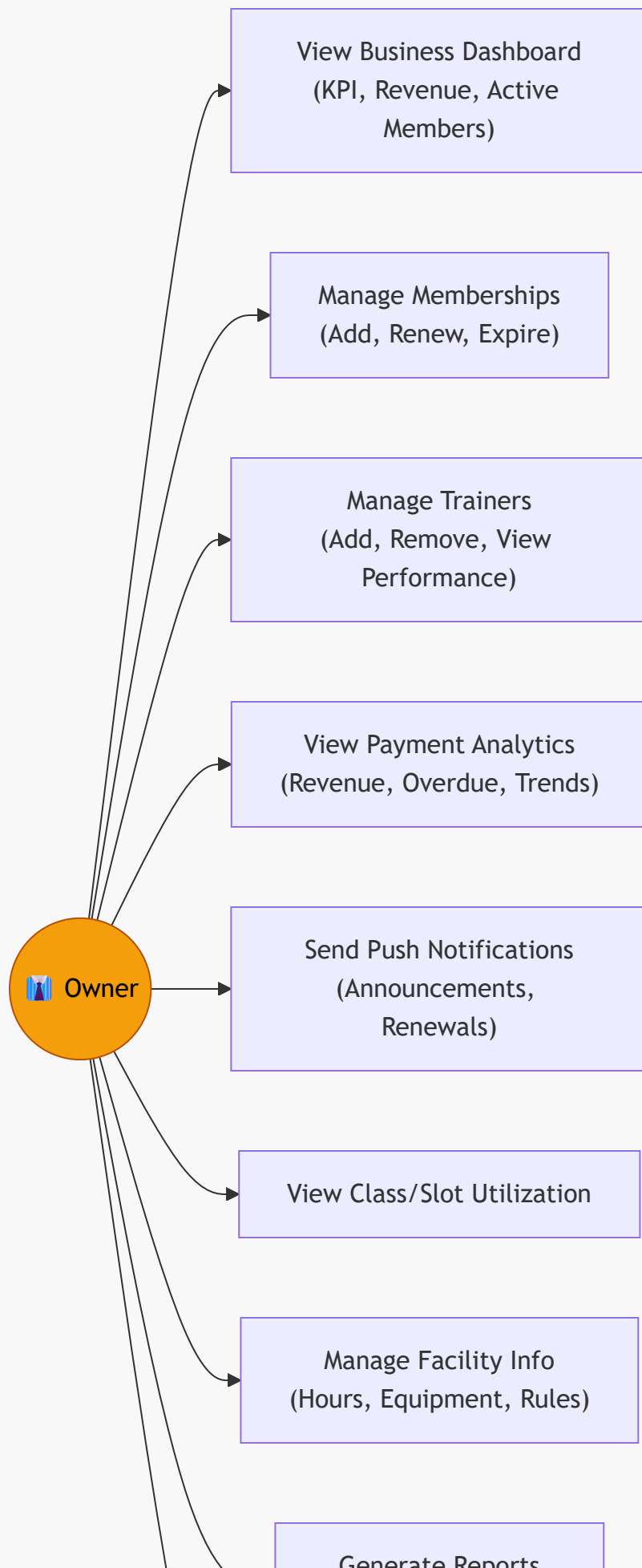


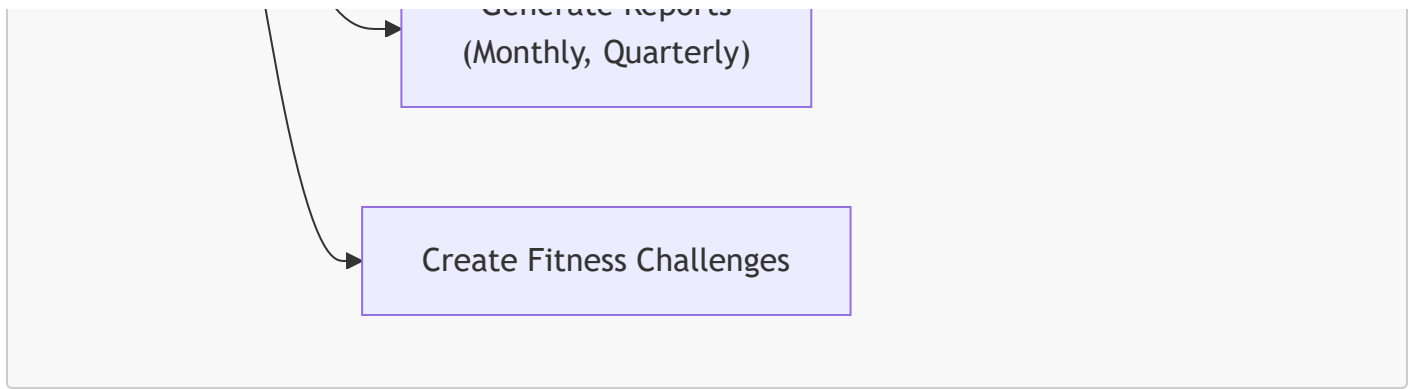
3. Data Model (Entity Relationship)



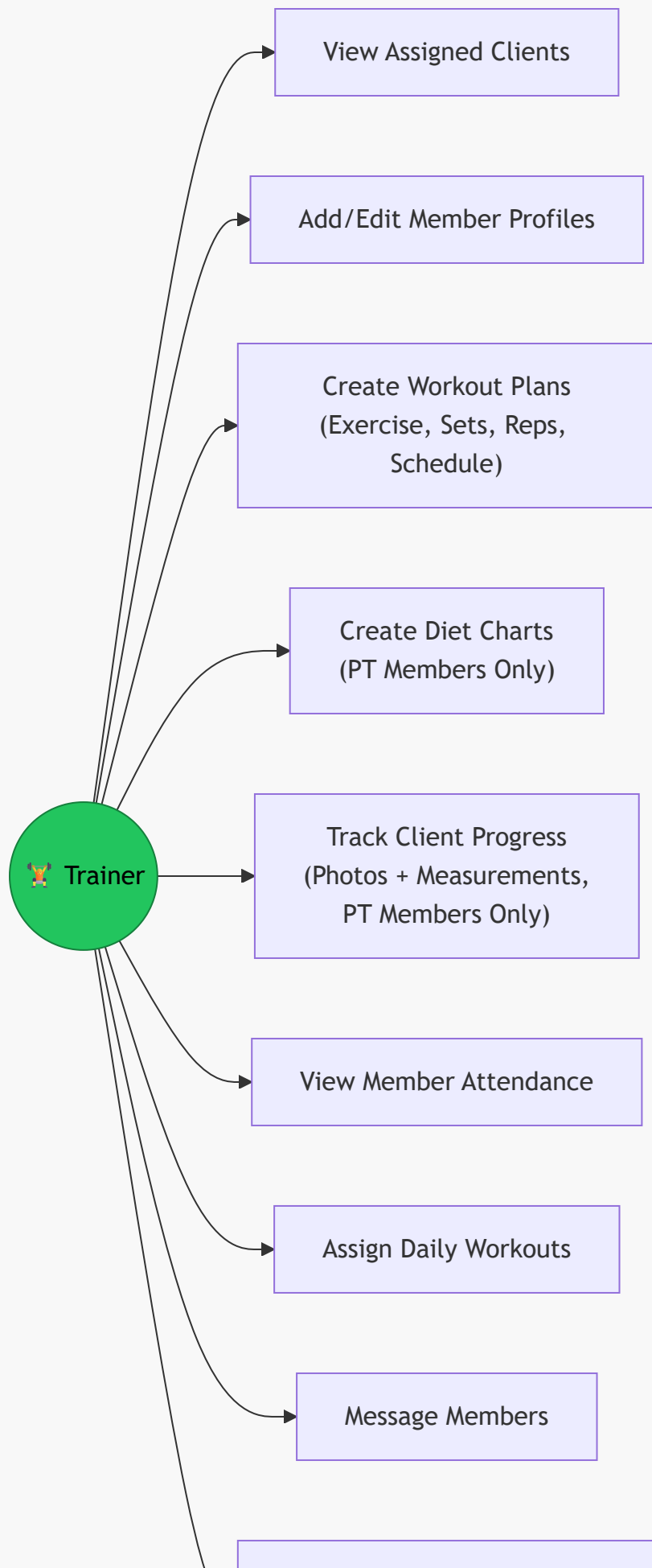
4. Use Case Diagrams

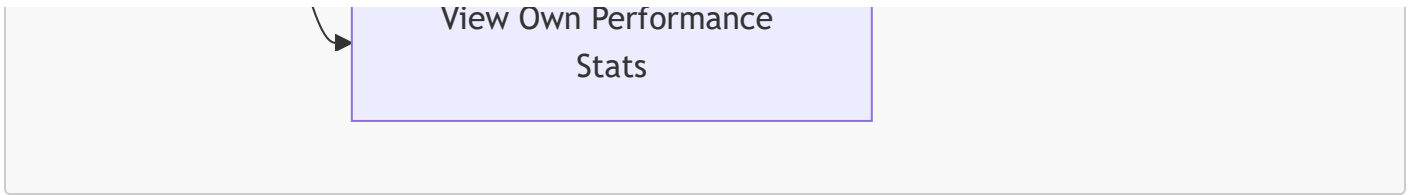
4.1 Owner Use Cases



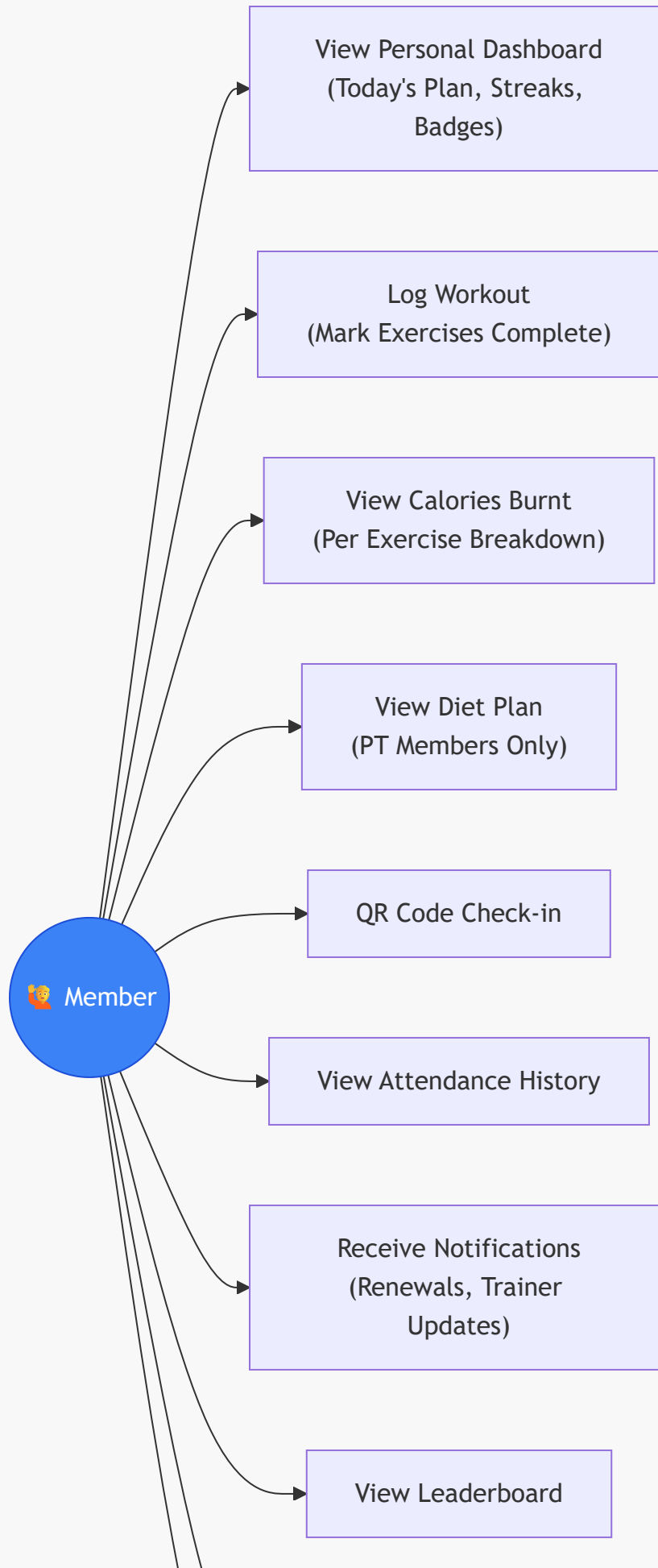


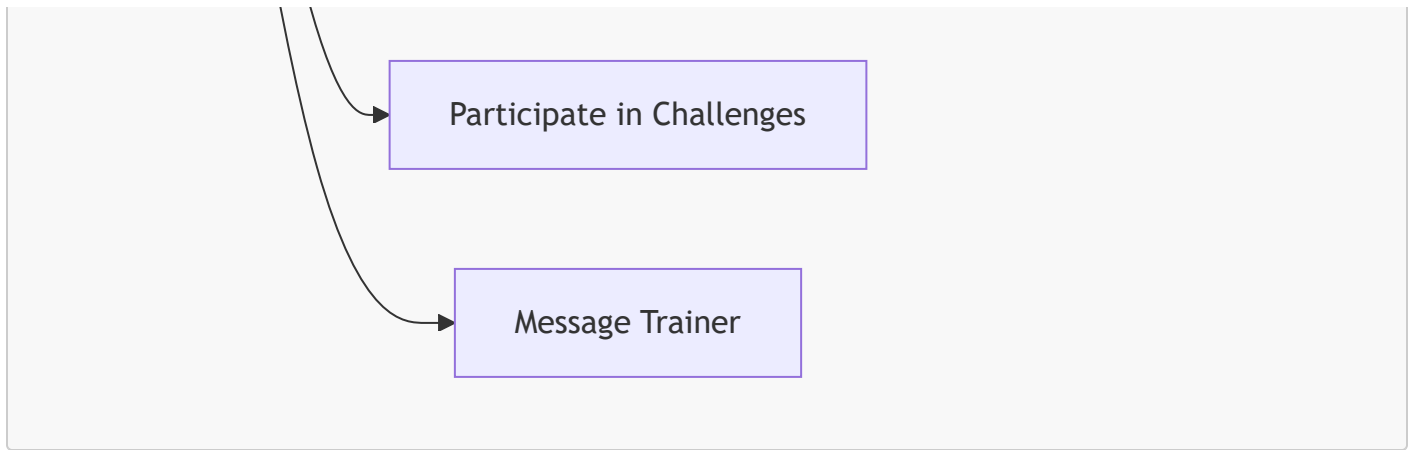
4.2 Trainer Use Cases





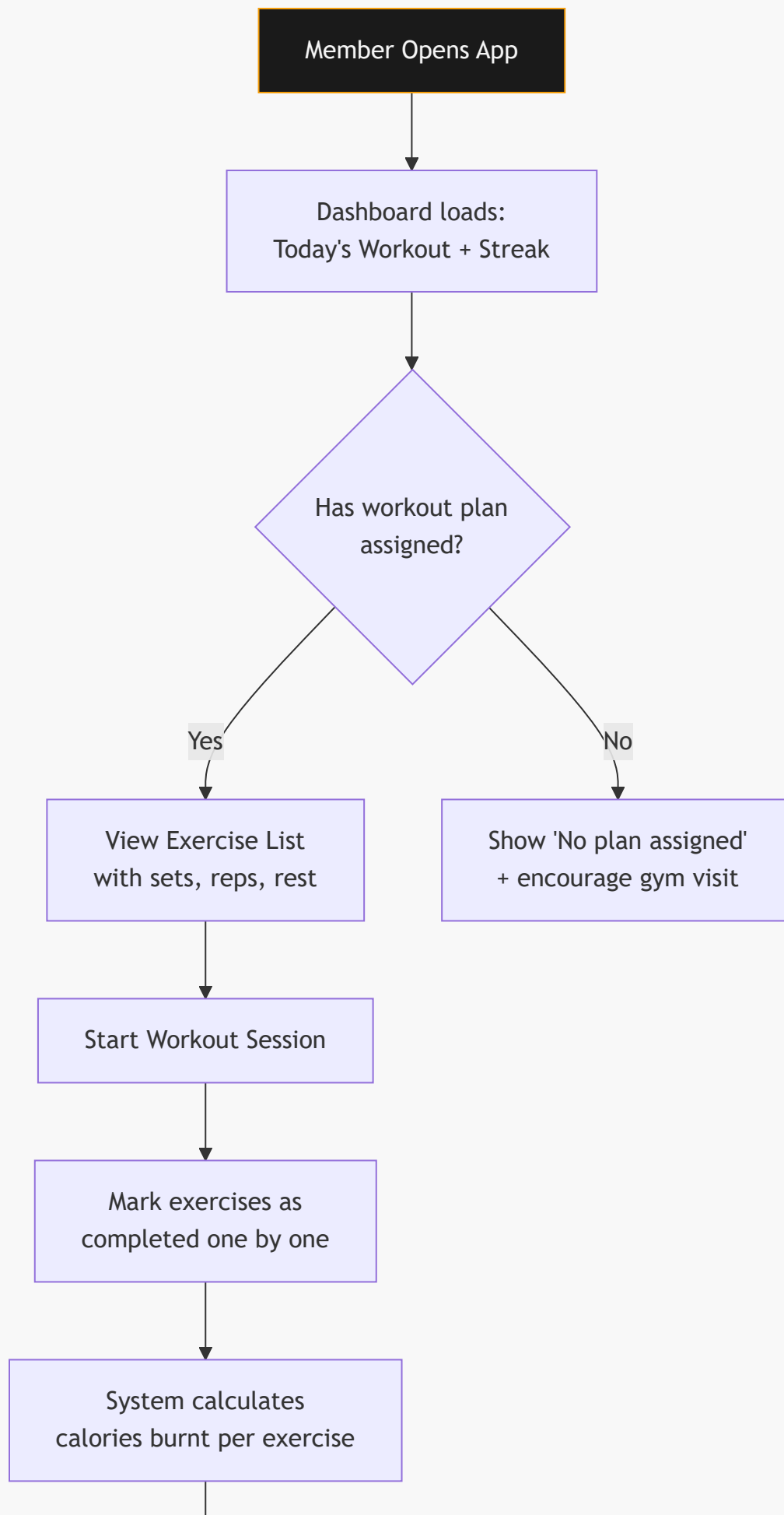
4.3 Member Use Cases

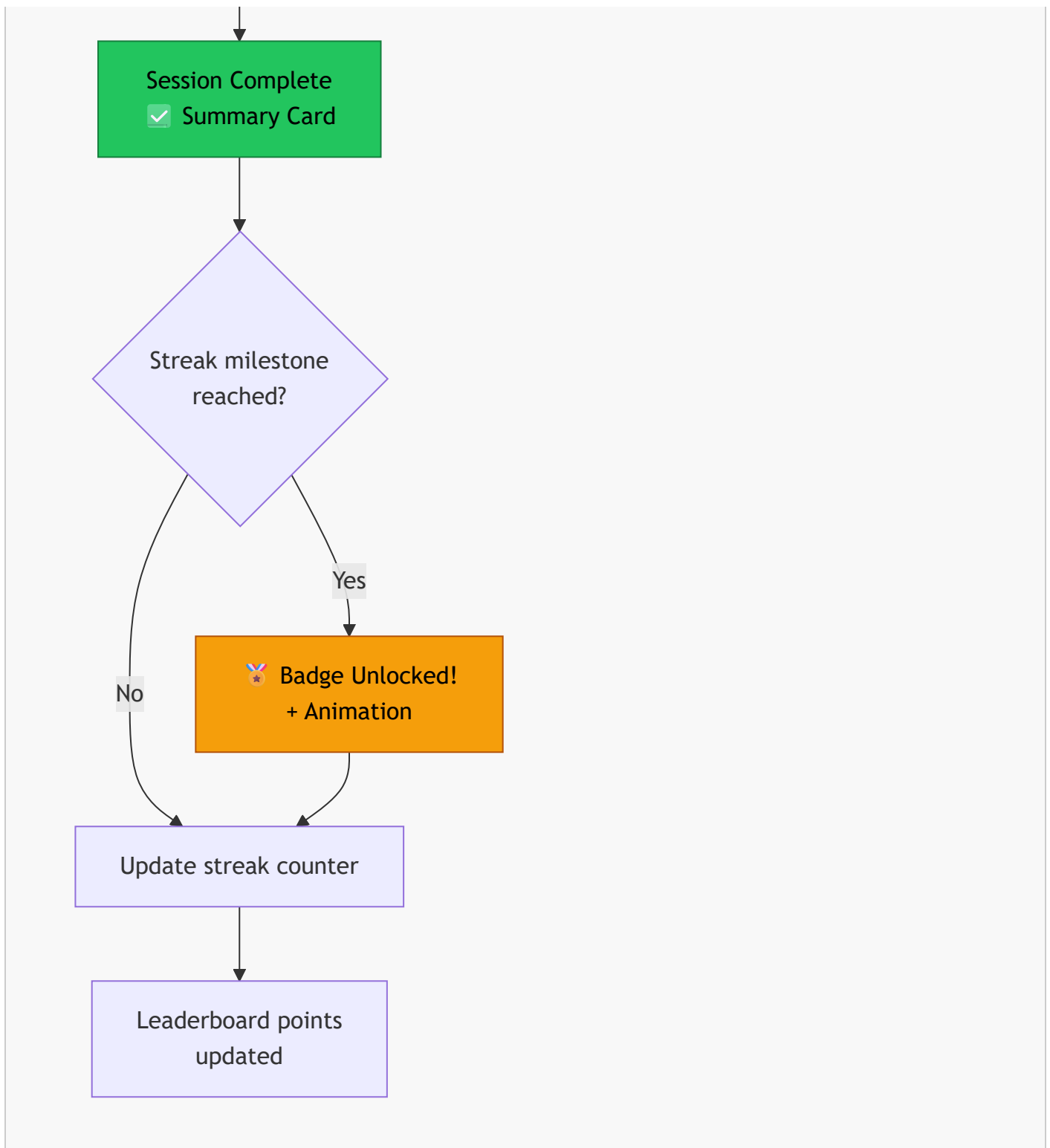




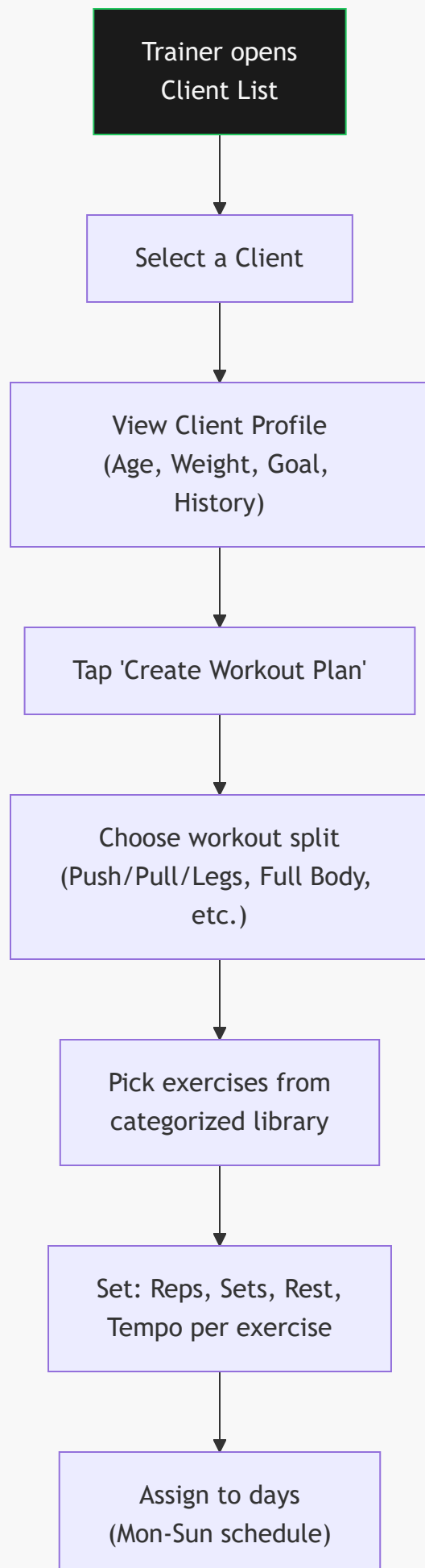
5. Core User Flows

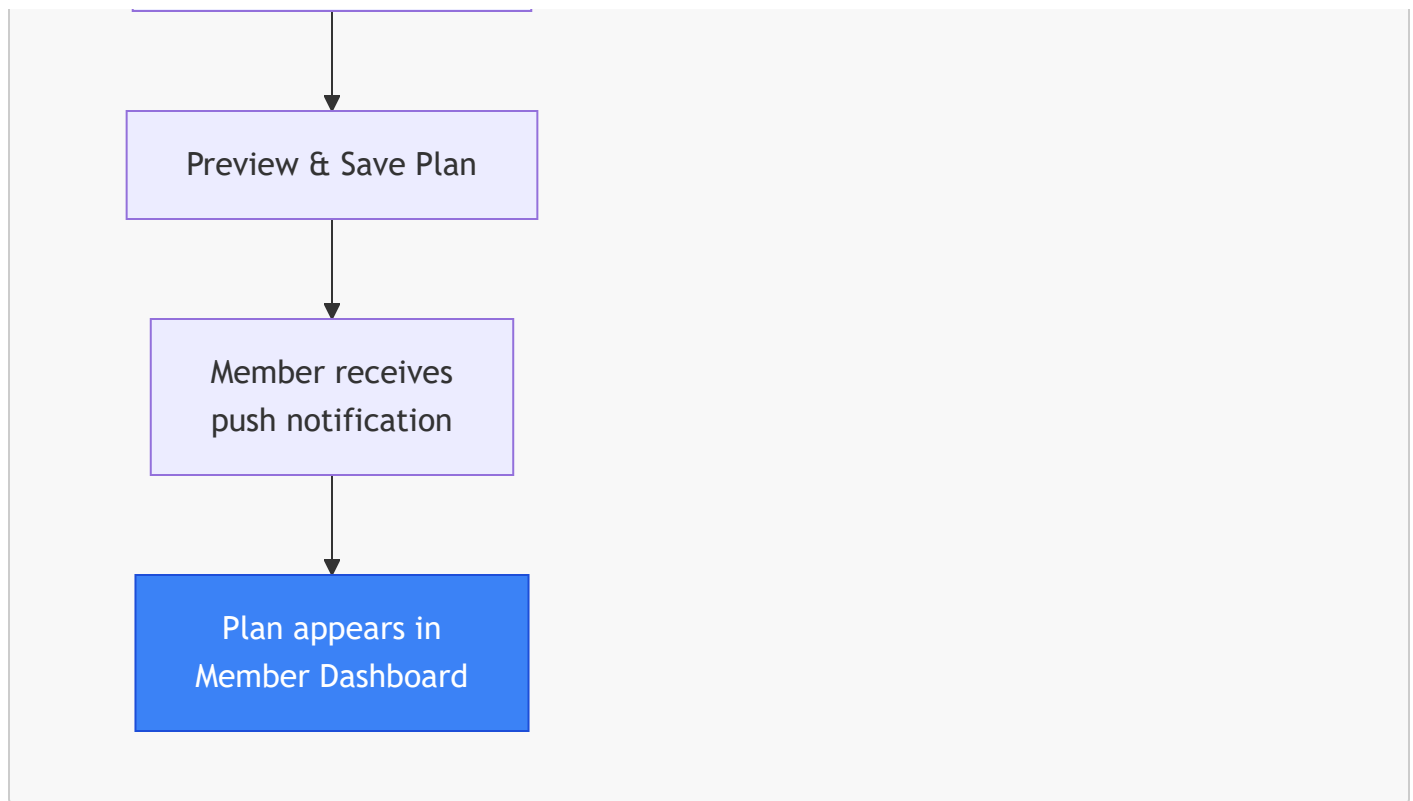
5.1 Member Daily Workout Flow



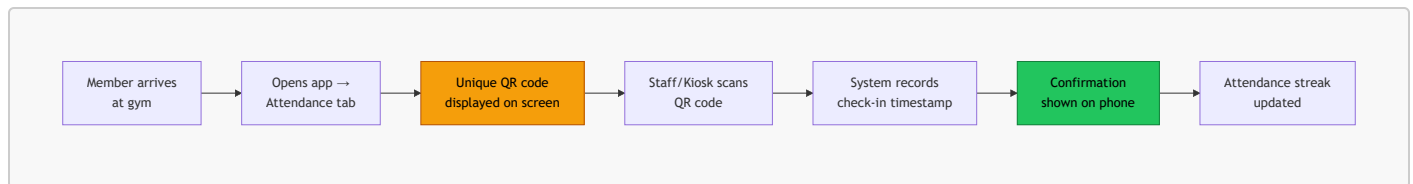


5.2 Trainer Creating a Workout Plan

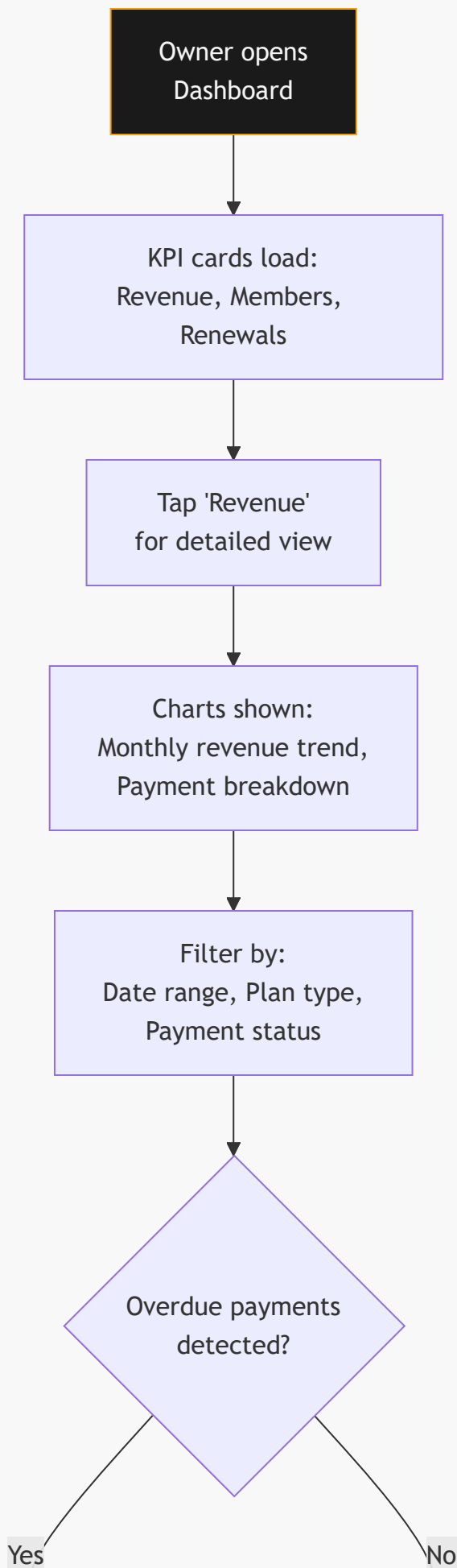


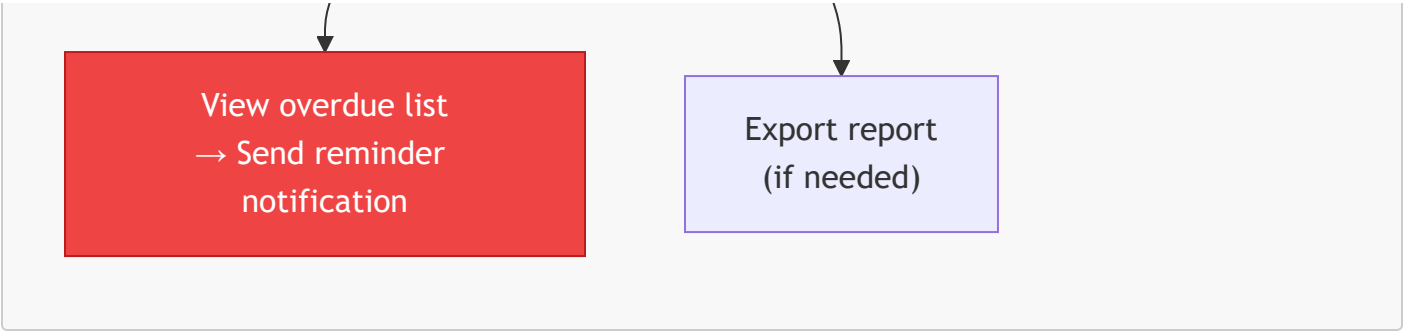


5.3 QR Attendance Check-in Flow



5.4 Owner Revenue Analytics Flow

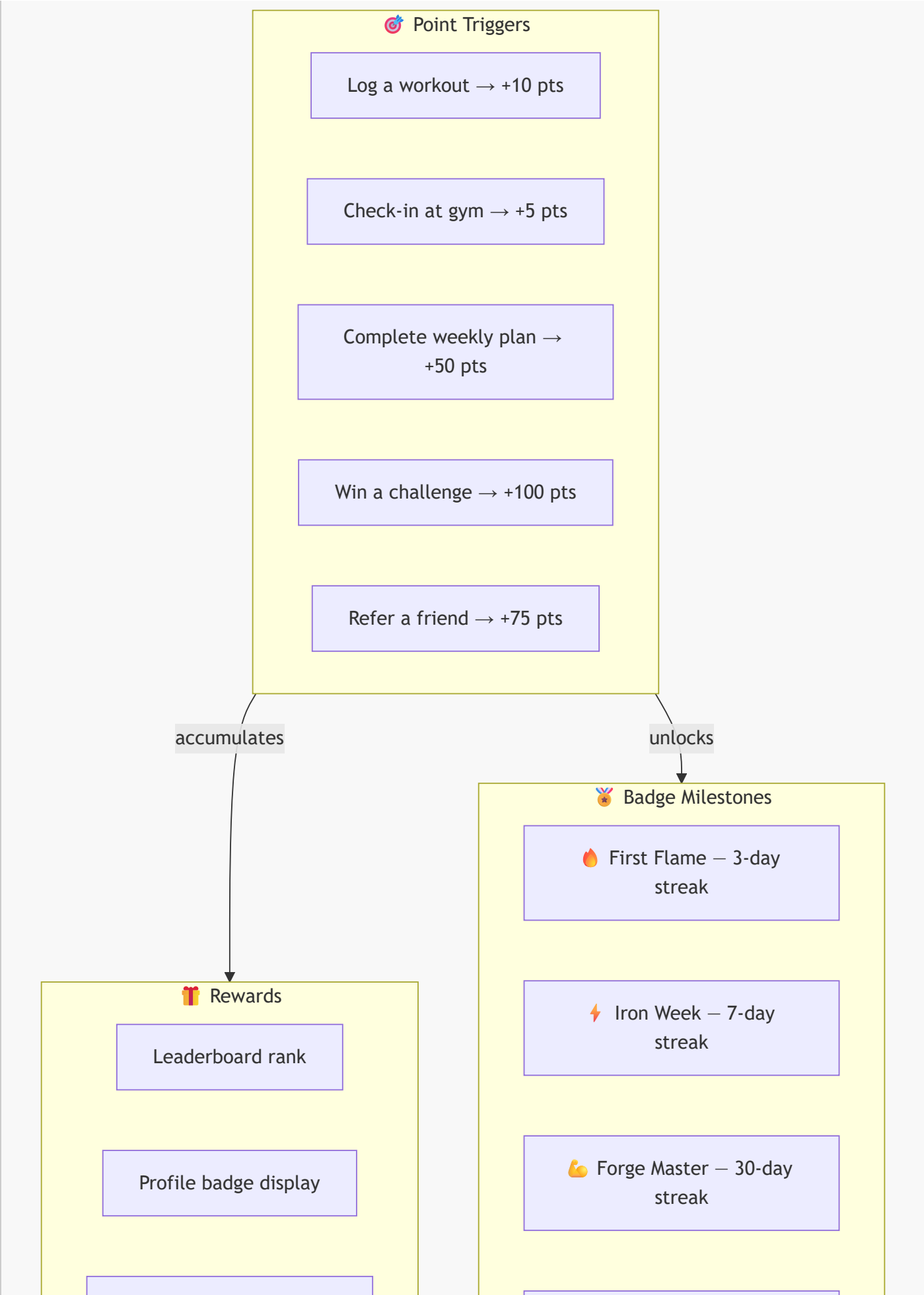


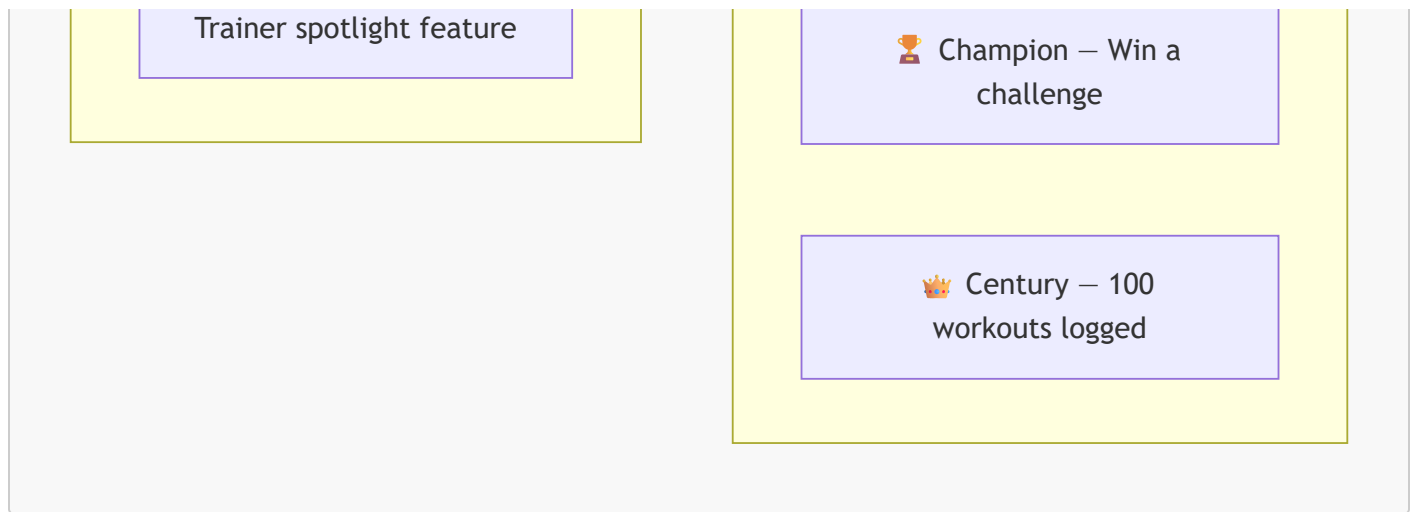


6. Module Feature Matrix

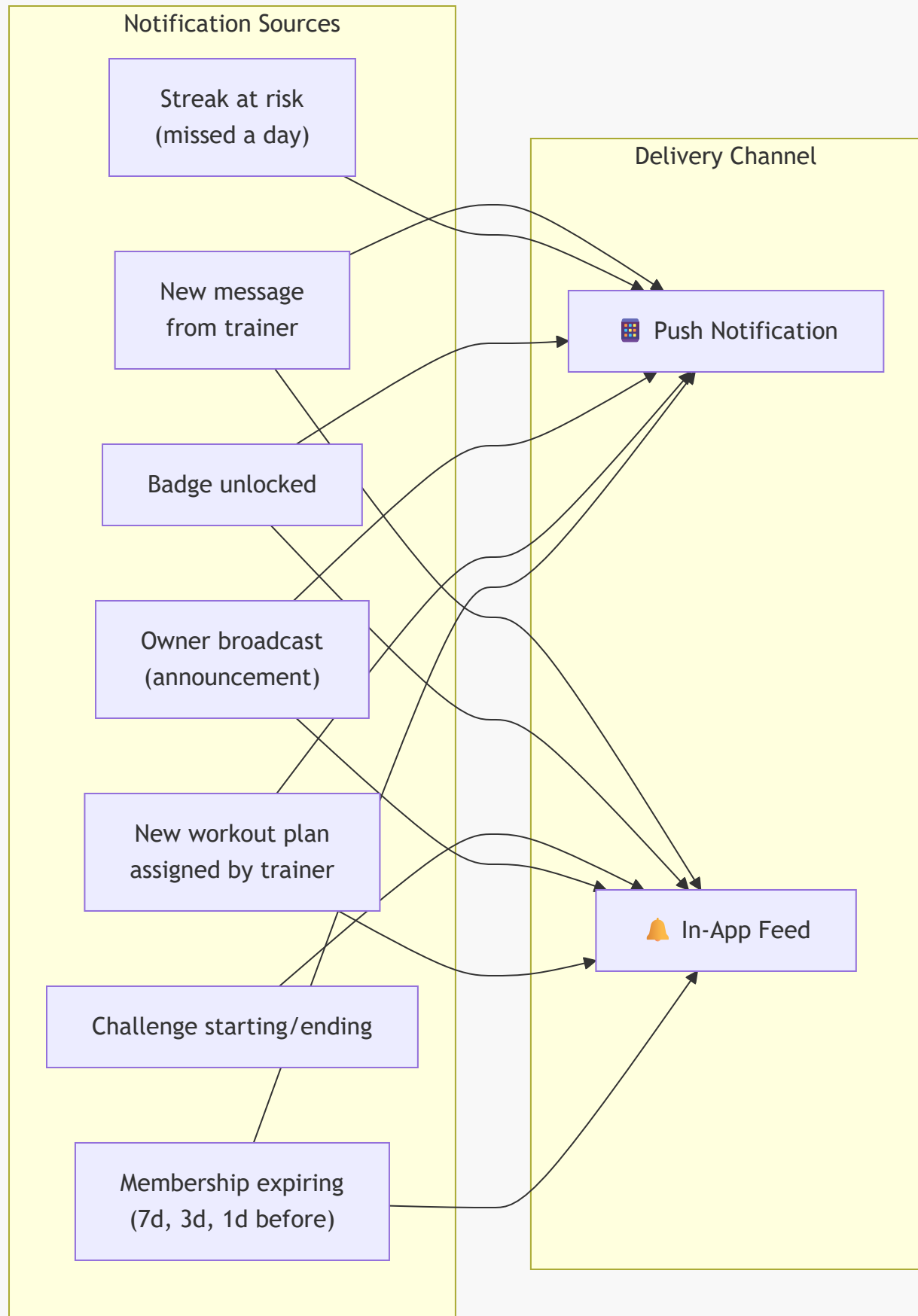
Feature	Owner	Trainer	Member (Regular)	Member (PT)
Business Dashboard	✓	—	—	—
Member Directory	✓	✓ (assigned only)	—	—
Trainer Directory	✓	—	—	—
Payment Analytics	✓	—	—	—
Push Notifications (Send)	✓	—	—	—
Facility Management	✓	—	—	—
Workout Plan Builder	—	✓	—	—
Diet Chart Builder	—	✓ (PT clients)	—	—
Progress Tracker	—	✓ (PT clients)	—	—
Client Attendance View	—	✓	—	—
Personal Dashboard	—	—	✓	✓
Workout Log	—	—	✓	✓
Calories Burnt Card	—	—	✓	✓
Diet Plan Viewer	—	—	—	✓
QR Check-in	—	—	✓	✓
Attendance History	—	—	✓	✓
Leaderboard	—	—	✓	✓
Challenges	—	—	✓	✓
Streaks & Badges	—	—	✓	✓
In-App Messaging	—	✓	✓	✓
Notifications (Receive)	—	✓	✓	✓

7. Gamification System





8. Notification System Map



9. Calorie Estimation Logic

The **Calories Burnt** card uses the standard **MET (Metabolic Equivalent of Task)** formula:

$$\text{Calories} = \text{MET} \times \text{Weight (kg)} \times \text{Duration (hours)}$$

Exercise	MET Value	Example (75 kg, 30 min)
Bench Press	6.0	225 kcal
Squats	5.5	206 kcal
Deadlifts	6.0	225 kcal
Running (treadmill)	9.8	368 kcal
Cycling (stationary)	7.0	263 kcal
Yoga	3.0	113 kcal
Jump Rope	12.3	461 kcal
Plank (static)	3.8	143 kcal

This is calculated automatically per exercise when a member logs their workout, and summed into a daily total on the dashboard.

10. Deployment Topology (Future State)

